Center Backs

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Player counts

In this project, there are a total of 154 center backs considered, with

- 39 center backs in 12 teams in the English FA Women's Super League
- 36 center backs in 12 teams in the German Frauen Bundesliga
- 53 center backs in 16 teams in the Spanish Liga F
- 26 center backs in 10 teams in the Italian Serie A Women

Results

Results for 2 clusters

Breakdown of results by league

league	FA Women's Super League	Frauen Bundesliga	Liga F	Serie A Women
kmeans_2_cluster				
0	23	24	36	18
1	16	12	17	8

Eventually the split between 2 clusters stay close to 2:1 for all four leagues.

Breakdown of results by features

- Cluster 0 center backs' defensive actions happen lower, while they passed more on the ground and their passing lengths under pressure were less affected.
- On the other hand, cluster 1 center backs engage more highly, both in terms of receiving and defending, while attempting more forward and progressive passes into the final third.

Therefore, based on these results,

- Cluster 0 center backs can be seen as traditional defenders who tend to hold the defensive line.
- Cluster 1 center backs can be seen as stoppers who are more aggressive in stepping up for defensive actions.

Results for 3 clusters

Breakdown of results by league

league	FA Women's Super League	Frauen Bundesliga	Liga F	Serie A Women
kmeans_3_cluster				
0	11	8	13	5
1	14	15	19	12
2	14	13	21	9

There seem to be more variances in terms of how players are splitted between clusters within each league.

Comparing results for 2 clusters against 3 clusters

kmeans_3_cluster	0	1	2
kmeans_2_cluster			
0	2	42	57
1	35	18	0

It seems like k = 3 cluster 1 is a shared middle zone between the two k = 2 clusters.

Breakdown of results by features

- Cluster 0 shows more positiveness towards average positioning for receiving and defensive actions, and progressive forward passing.
- Cluster 2, on the other hand, shows more positiveness towards average pass length change under pressure, while engage in defensive actions at much lower positions.
- While for cluster 1, most of their features stayed in between the other two clusters, not much really stood out.

Hence, the following conclusions can be made:

- Similar to k = 2 cluster 0, k = 3 cluster 2 center backs can be seen as traditional defenders holding the line.
- Similar to k = 2 cluster 1, k = 3 cluster 0 center backs can be seen as stoppers.
- Cluster 1, while they engage in defending as highly as cluster 0, their passing look close to cluster 2, which can be seen as high engagement low progressing center backs.

However, these results are not stable as number of components are added, so results are not highly confident.

Breakdown of results by players

Findings and radar plots for each center back are in their own files:

- Molly Bartrip - Tottenham Hotspur
- Elba Verges Prats - SD Eibar
- Kathy Hendrich - VfL Wolfsburg
- Sophie Howard - Leicester City
- Risa Shimizu - West Ham United
- Alex Greenwood - Manchester City

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Appendix

Radar plots for 2 clusters center backs' averages

All leagues

Passing

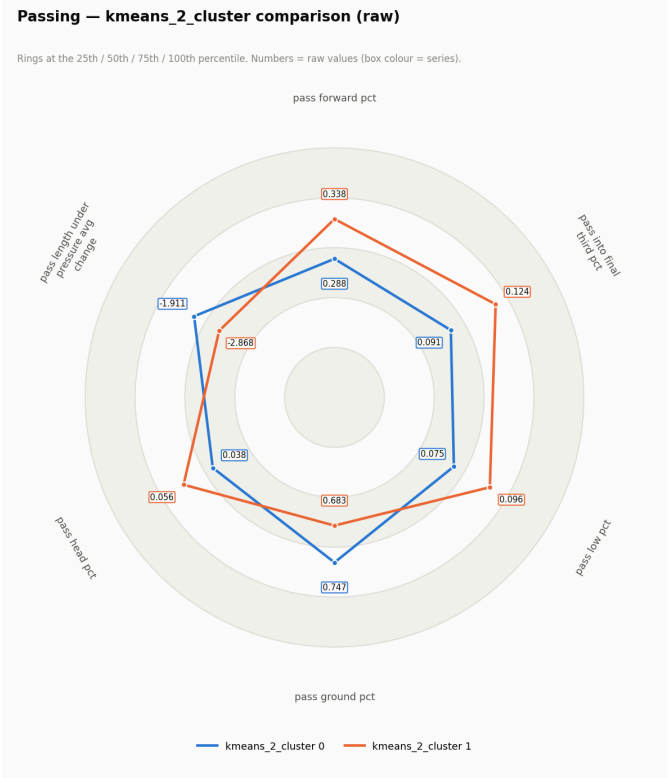


Figure 1: Center Backs Passing — Raw Statistics Comparison, 2 Clusters

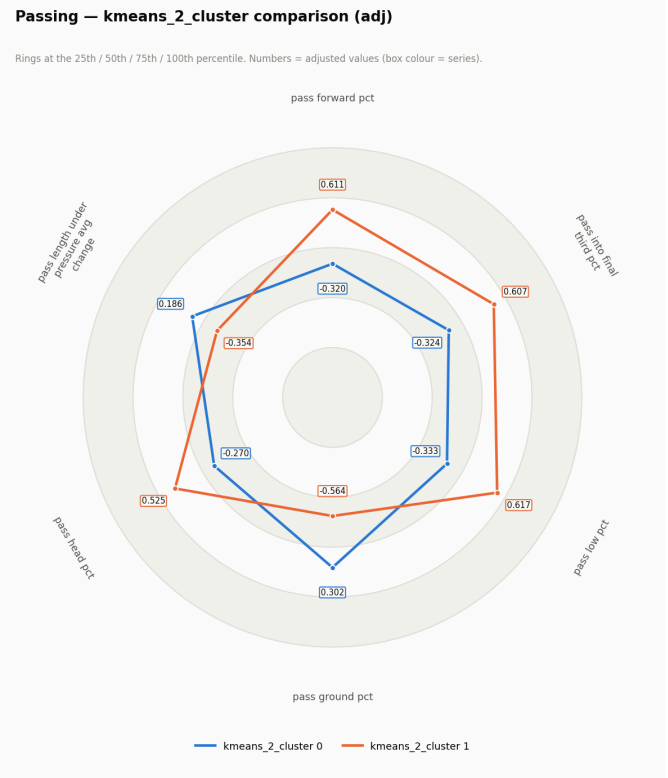


Figure 2: Center Backs Passing — Adjusted Statistics Comparison, 2 Clusters

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Carrying & Receipt

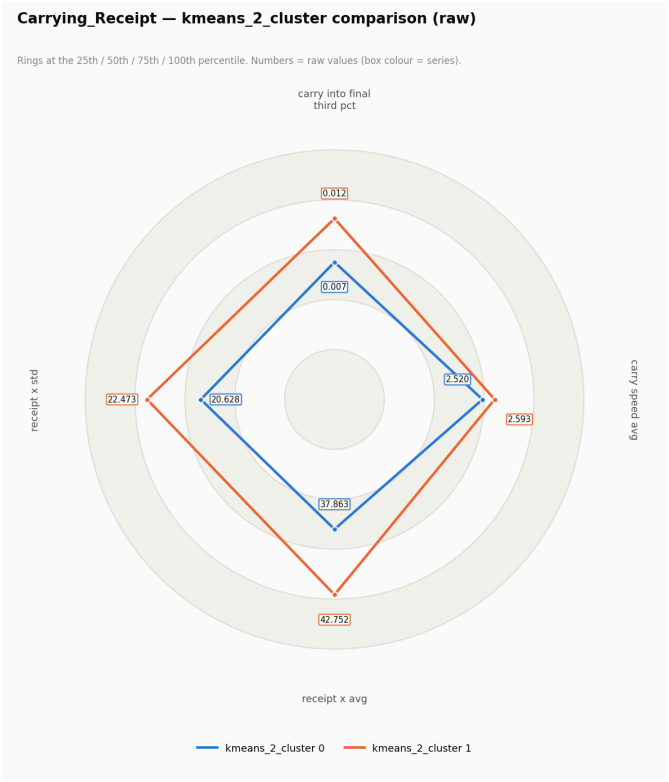


Figure 3: Center Backs Carrying Receipt — Raw Statistics Comparison, 2 Clusters

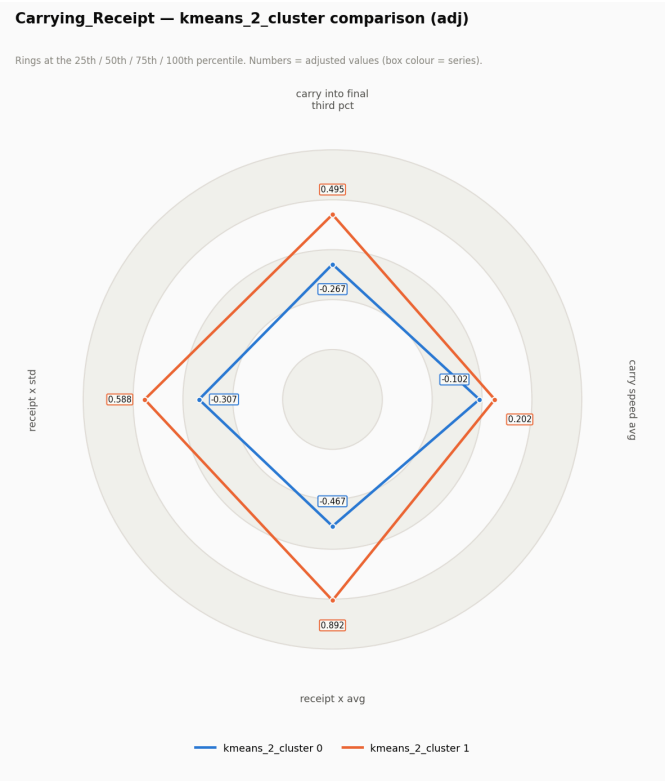


Figure 4: Center Backs Carrying Receipt — Adjusted Statistics Comparison, 2 Clusters

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Defending & Vulnerability

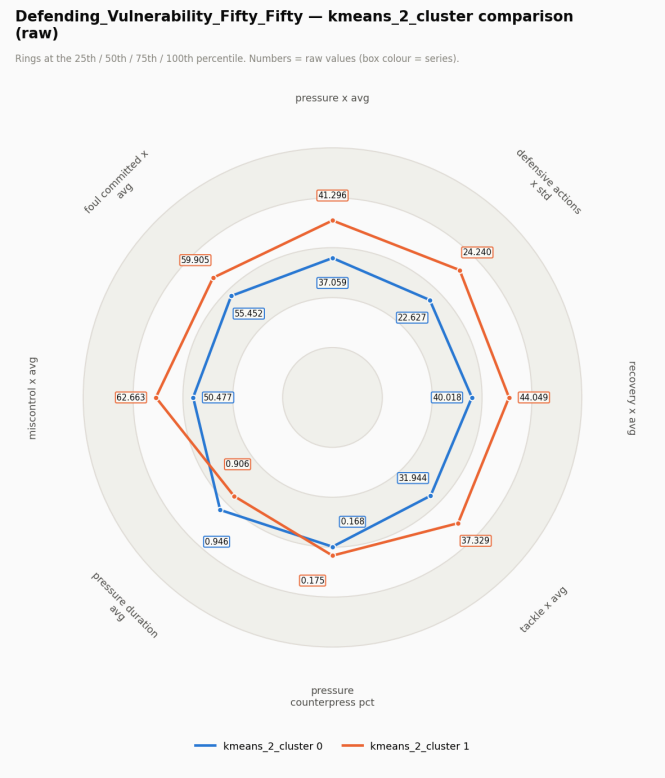


Figure 5: Center Backs Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 2 Clusters

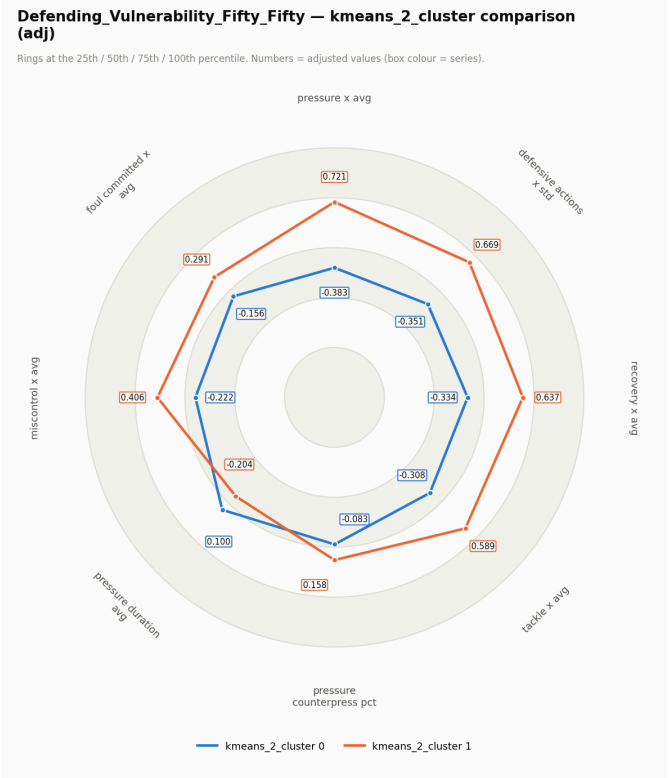


Figure 6: Center Backs Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 2 Clusters

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FA Women's Super League

Passing

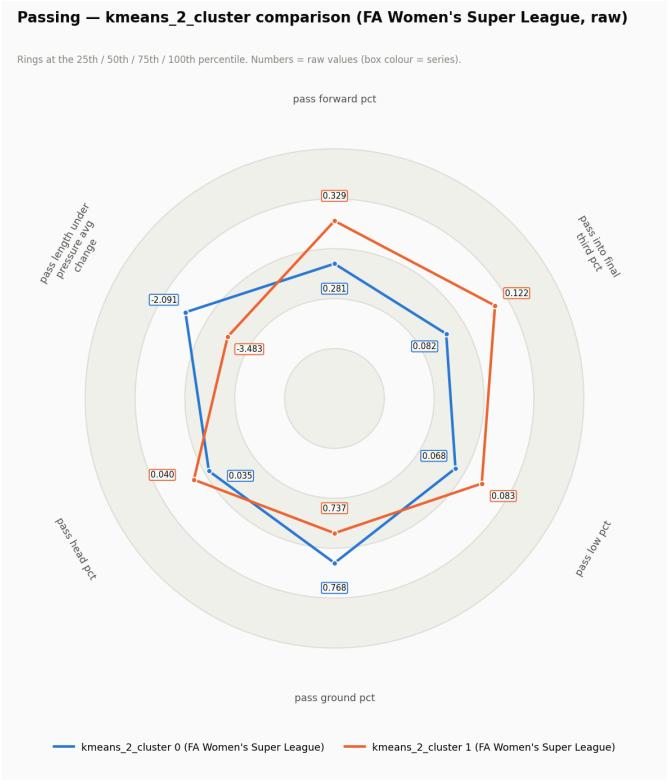


Figure 7: England Center Backs Passing — Raw Statistics Comparison, 2 Clusters

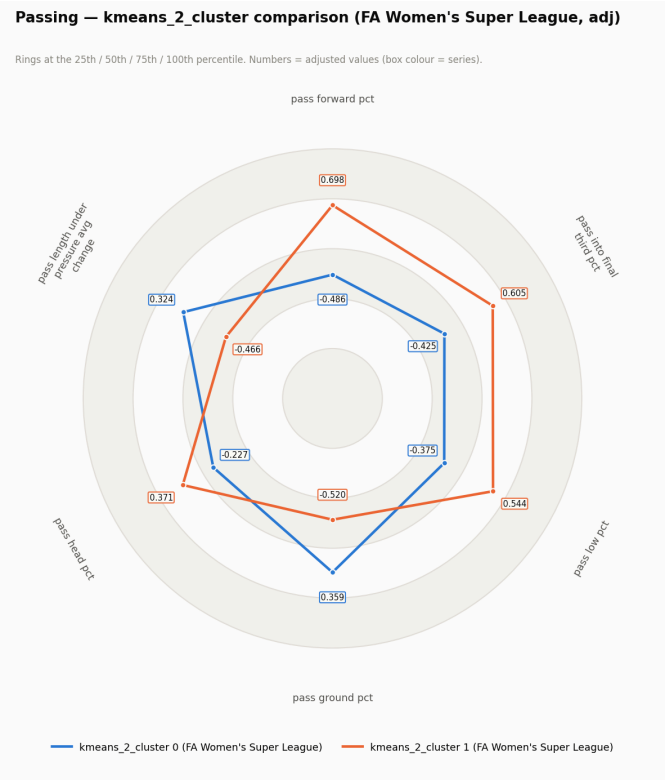


Figure 8: England Center Backs Passing — Adjusted Statistics Comparison, 2 Clusters

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Carrying & Receipt

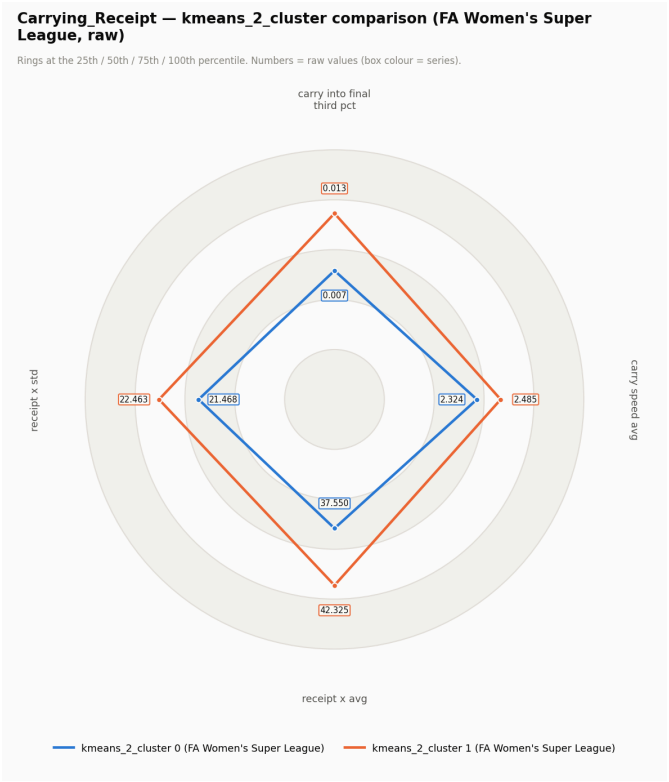


Figure 9: England Center Backs Carrying Receipt — Raw Statistics Comparison, 2 Clusters

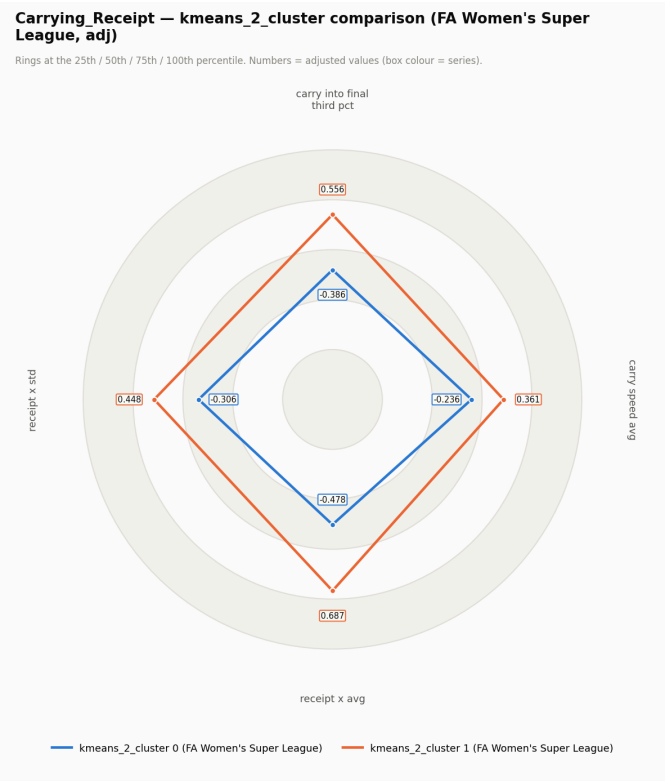


Figure 10: England Center Backs Carrying Receipt — Adjusted Statistics Comparison, 2 Clusters

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Defending & Vulnerability

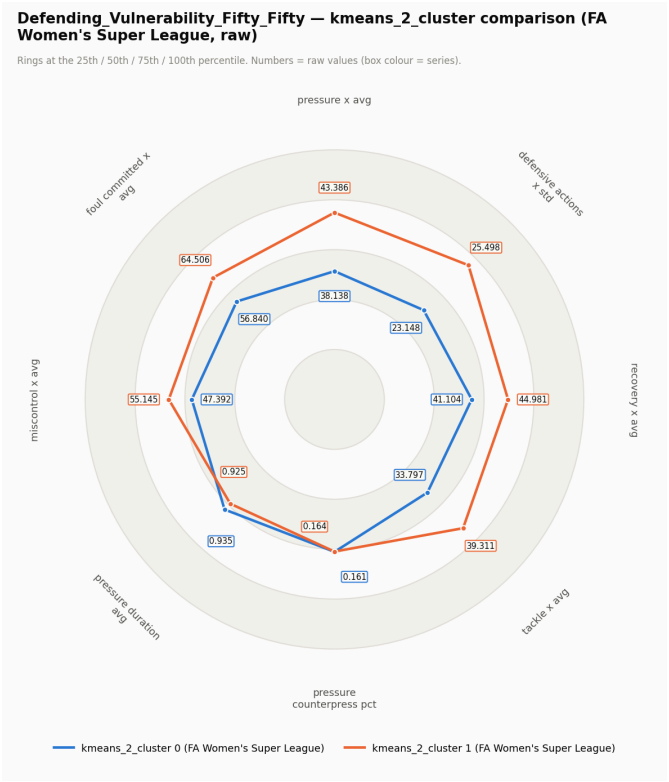


Figure 11: England Center Backs Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 2 Clusters

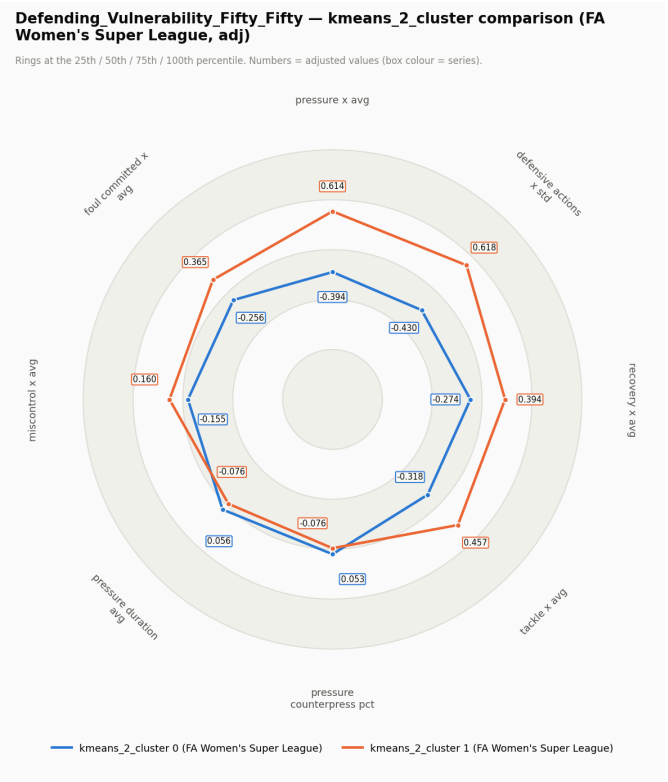


Figure 12: England Center Backs Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 2 Clusters

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Frauen Bundesliga

Passing

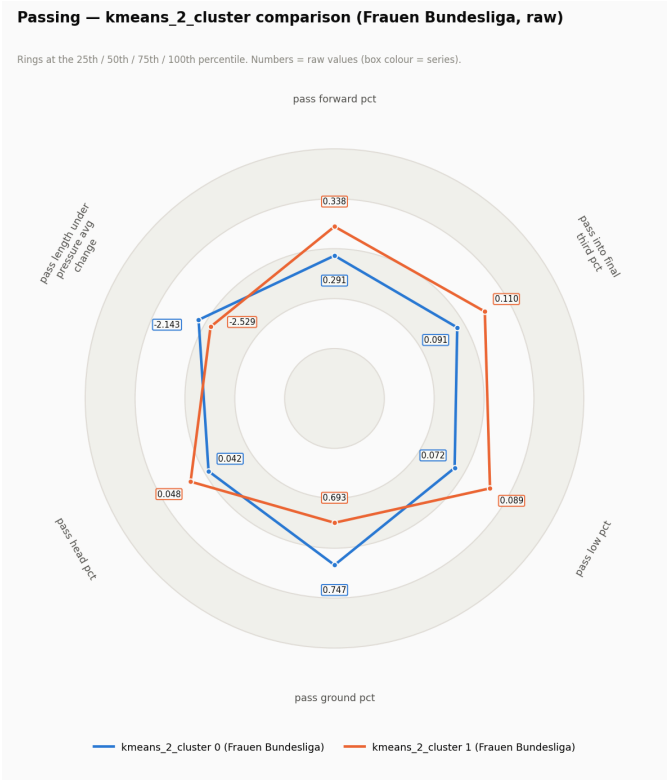


Figure 13: Germany Center Backs Passing — Raw Statistics Comparison, 2 Clusters

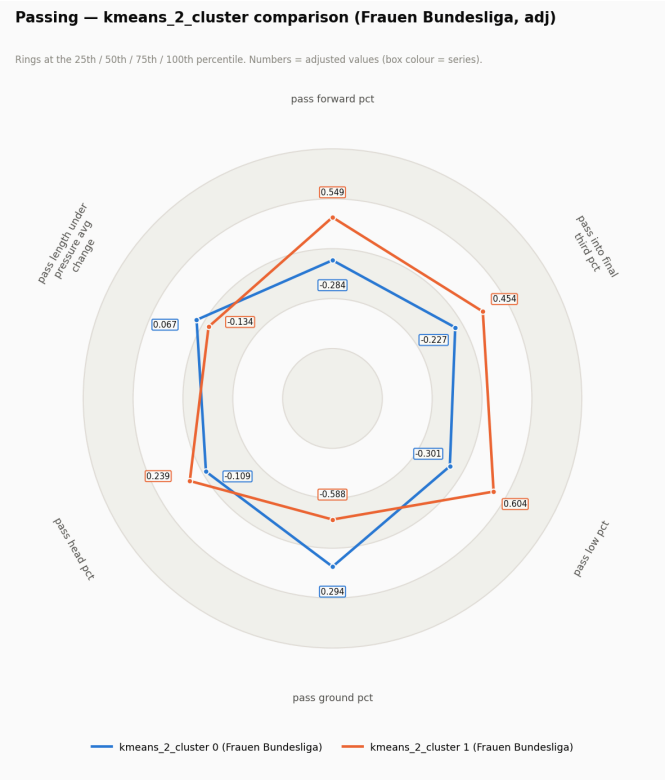


Figure 14: Germany Center Backs Passing — Adjusted Statistics Comparison, 2 Clusters

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Carrying & Receipt

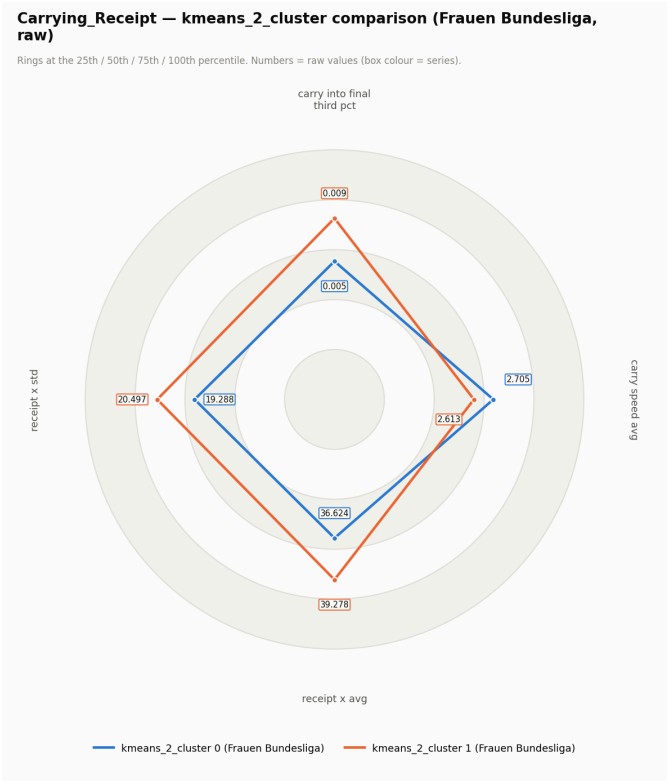


Figure 15: Germany Center Backs Carrying Receipt — Raw Statistics Comparison, 2 Clusters

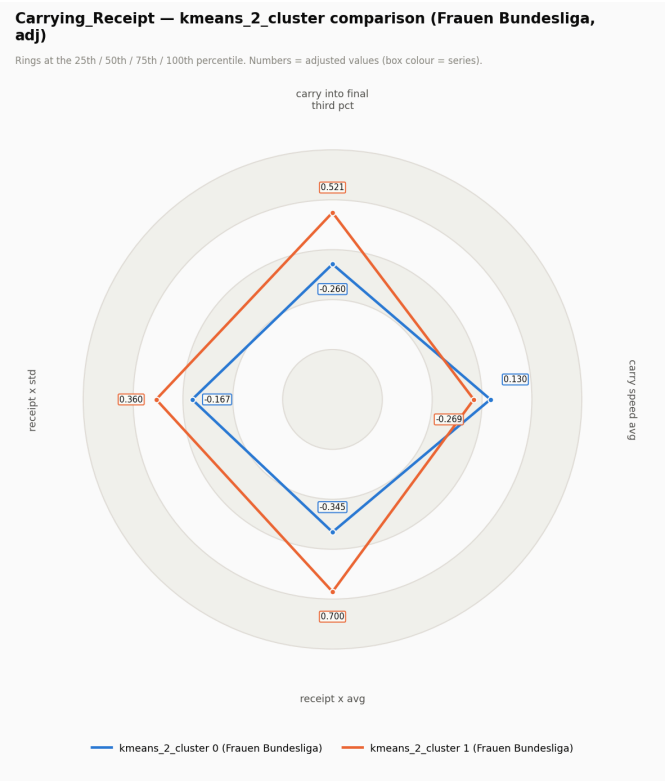


Figure 16: Germany Center Backs Carrying Receipt — Adjusted Statistics Comparison, 2 Clusters

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Defending & Vulnerability

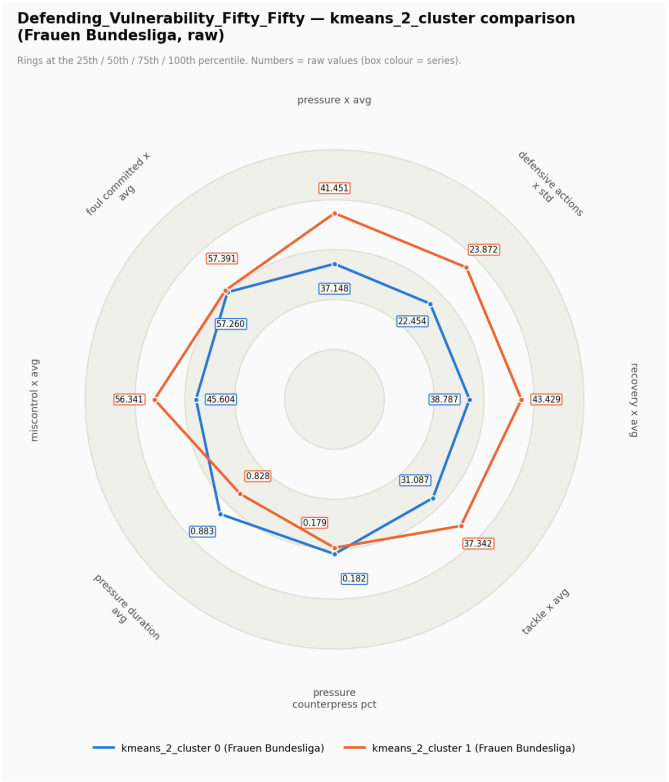


Figure 17: Germany Center Backs Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 2 Clusters

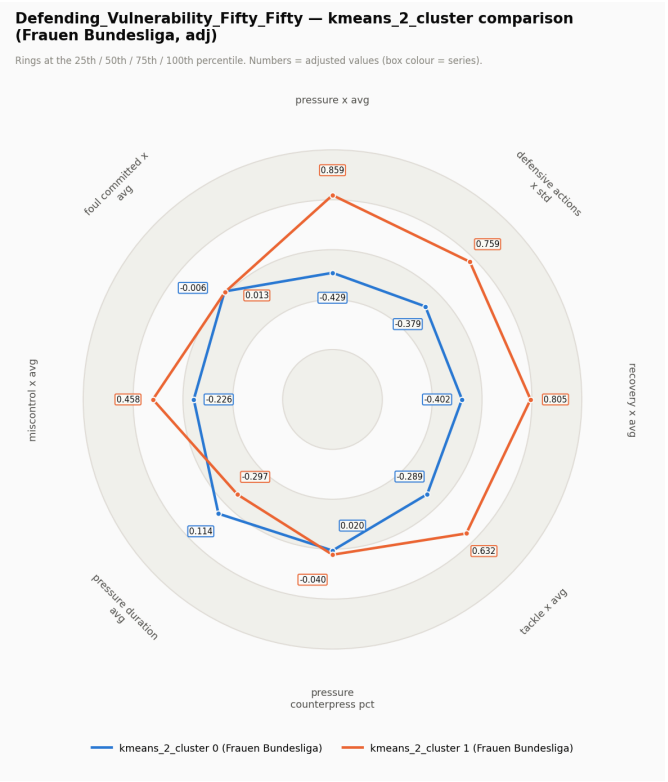


Figure 18: Germany Center Backs Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 2 Clusters

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Liga F

Passing

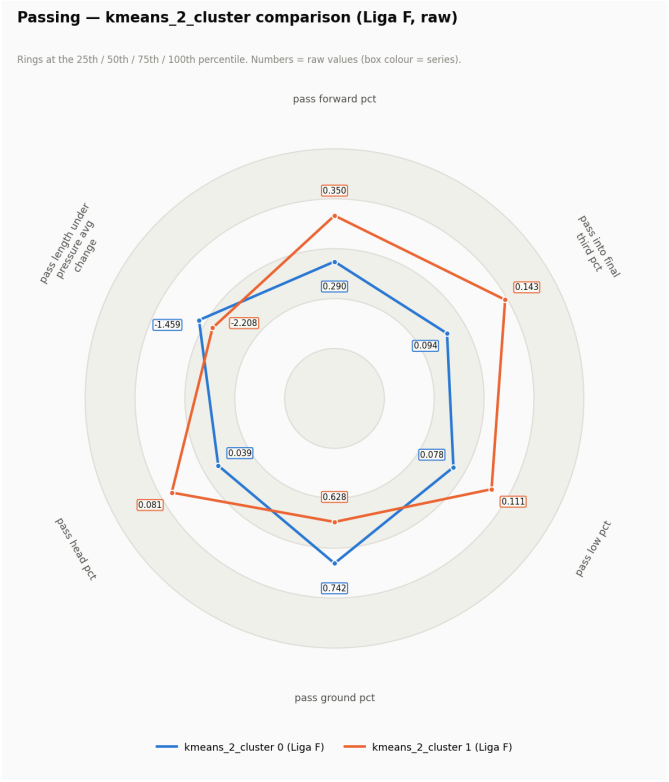


Figure 19: Spain Center Backs Passing — Raw Statistics Comparison, 2 Clusters

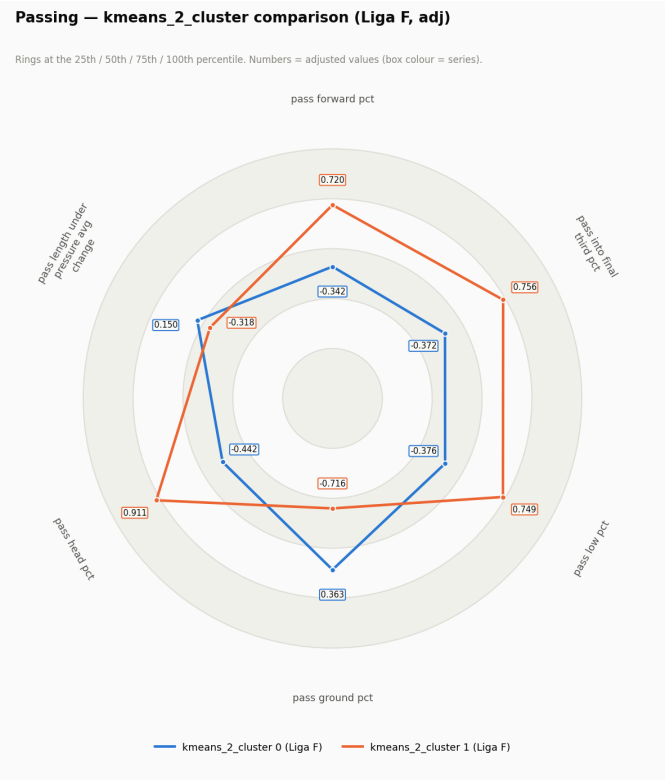


Figure 20: Spain Center Backs Passing — Adjusted Statistics Comparison, 2 Clusters

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Carrying & Receipt

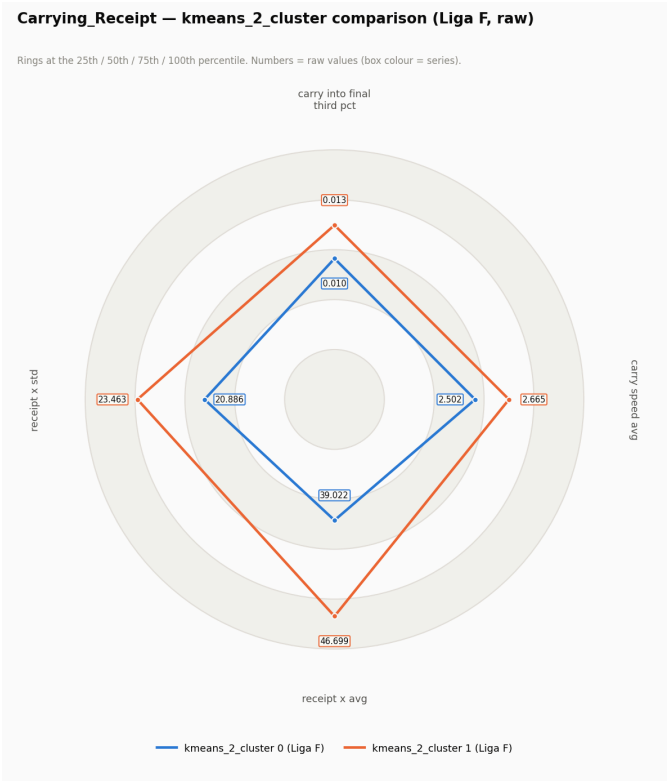


Figure 21: Spain Center Backs Carrying Receipt — Raw Statistics Comparison, 2 Clusters

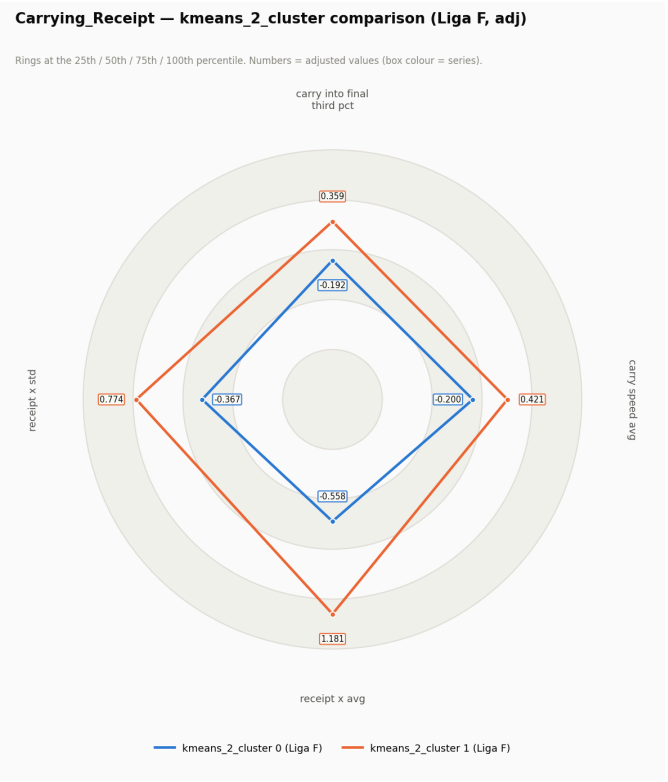


Figure 22: Spain Center Backs Carrying Receipt — Adjusted Statistics Comparison, 2 Clusters

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Defending & Vulnerability

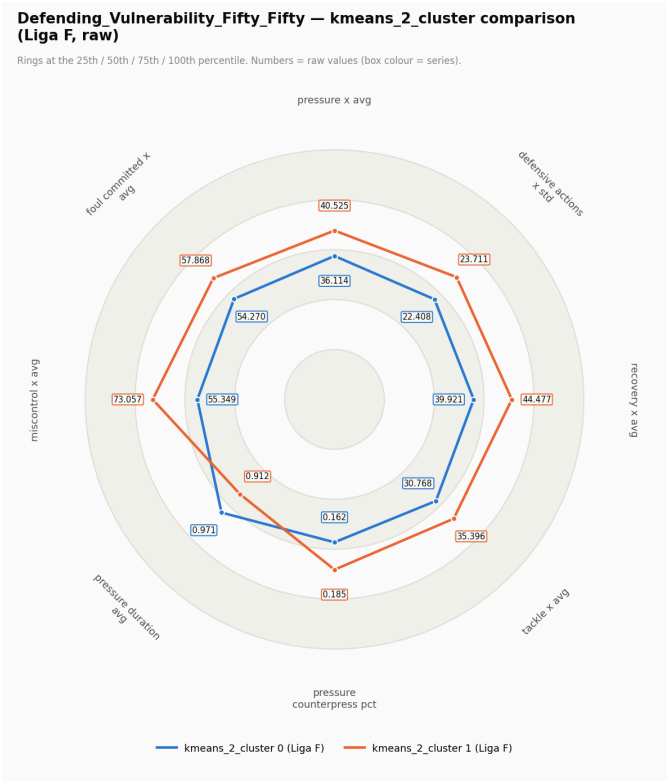


Figure 23: Spain Center Backs Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 2 Clusters

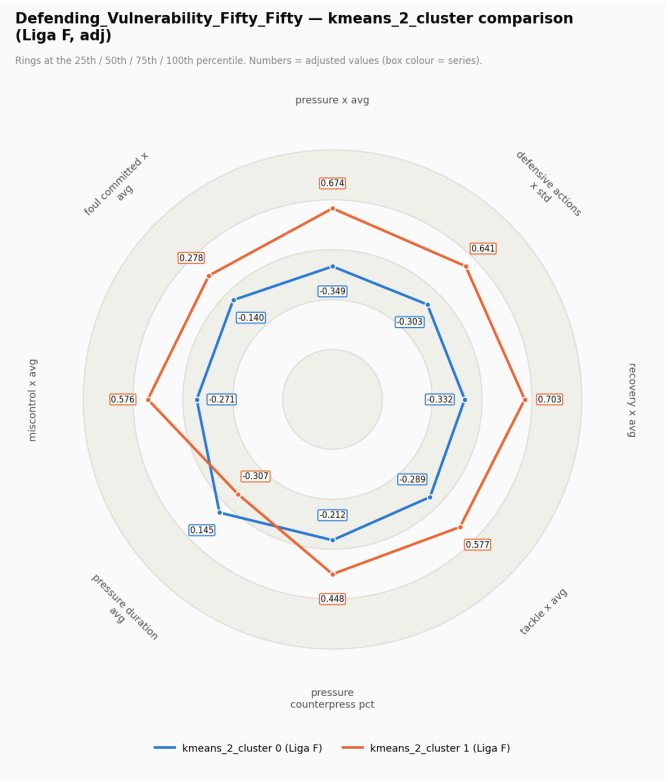


Figure 24: Spain Center Backs Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 2 Clusters

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Serie A Women

Passing

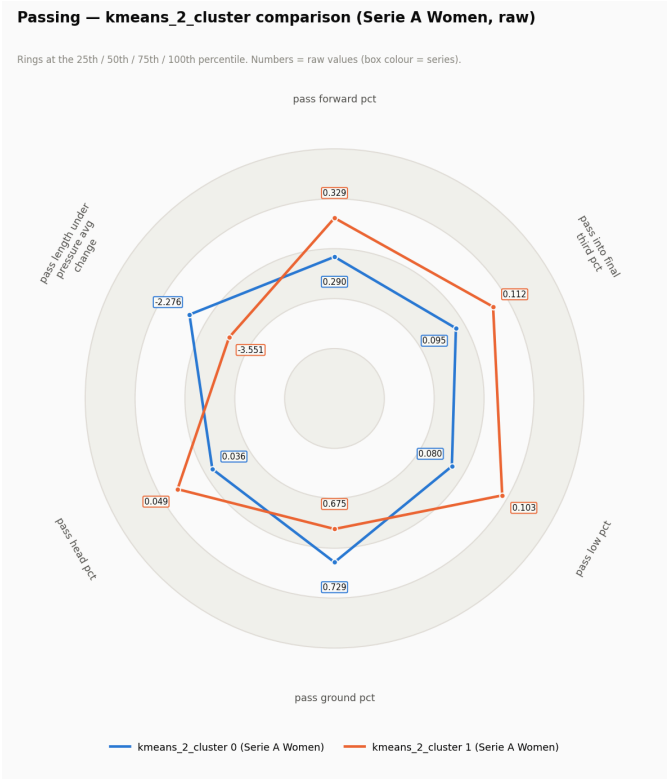


Figure 25: Italy Center Backs Passing — Raw Statistics Comparison, 2 Clusters

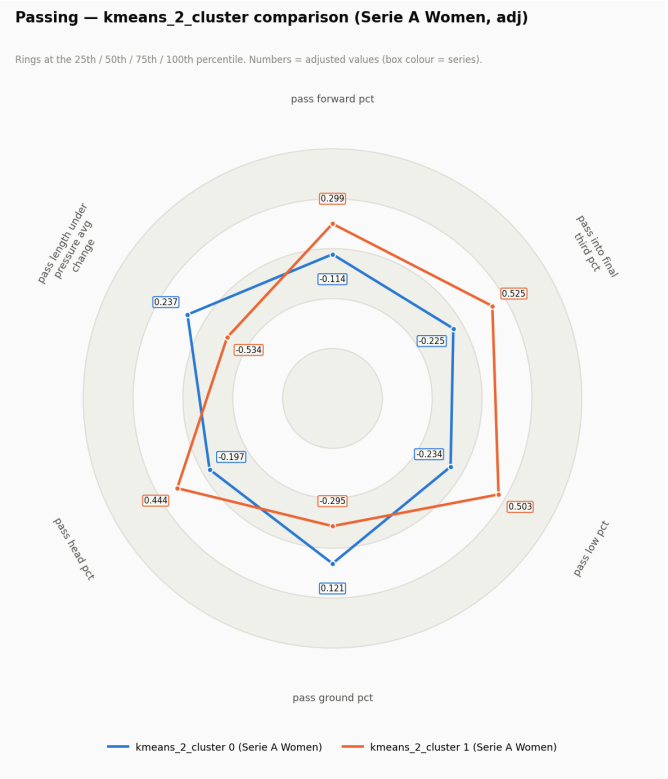


Figure 26: Italy Center Backs Passing — Adjusted Statistics Comparison, 2 Clusters

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Carrying & Receipt

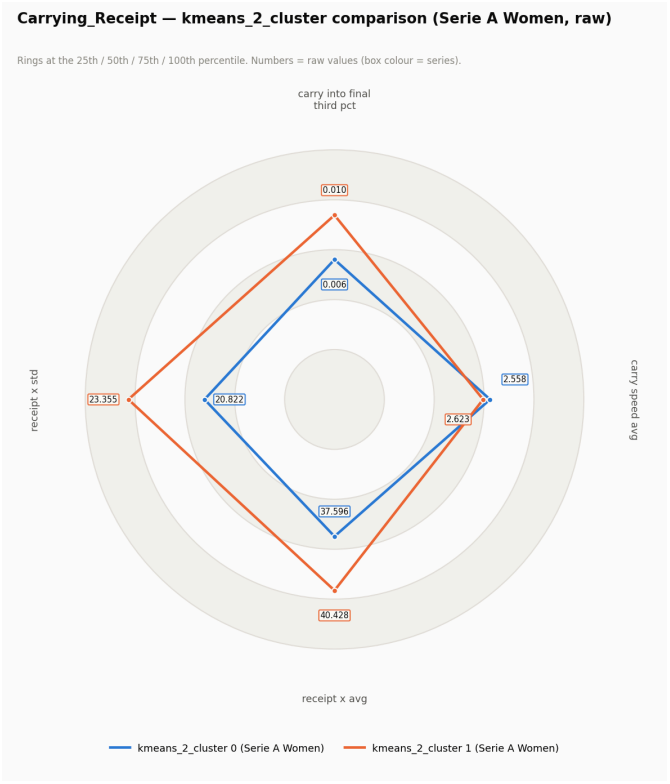


Figure 27: Italy Center Backs Carrying Receipt — Raw Statistics Comparison, 2 Clusters

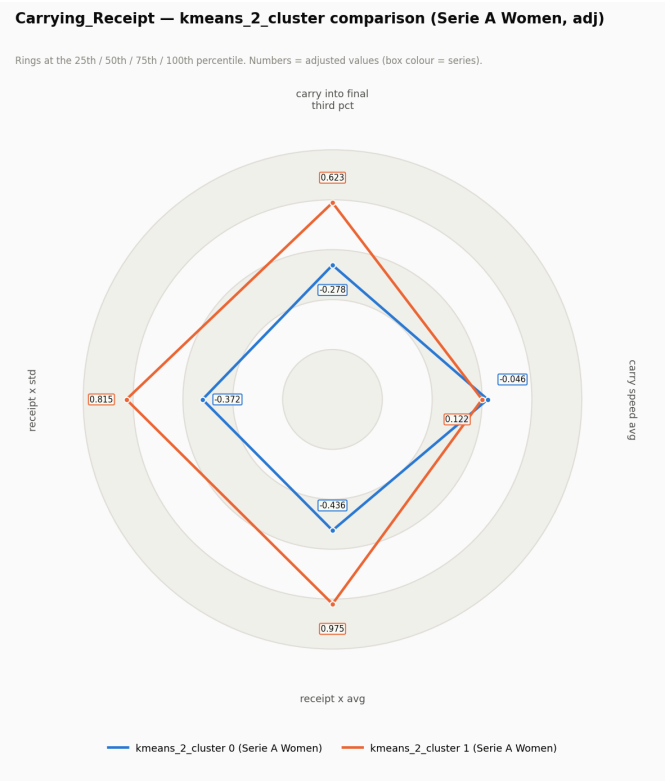


Figure 28: Italy Center Backs Carrying Receipt — Adjusted Statistics Comparison, 2 Clusters

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Defending & Vulnerability

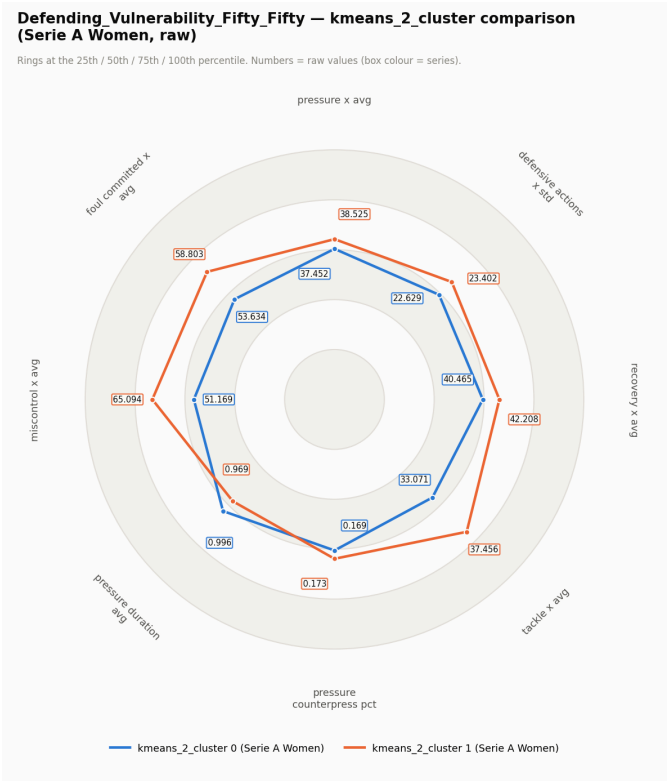


Figure 29: Italy Center Backs Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 2 Clusters

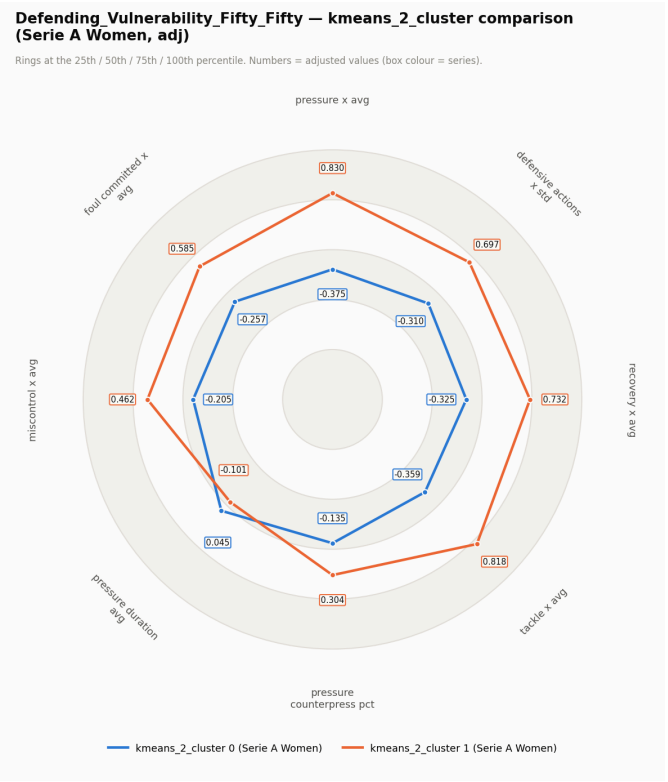


Figure 30: Italy Center Backs Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 2 Clusters

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Radar plots for 3 clusters center backs' averages

All leagues

Passing

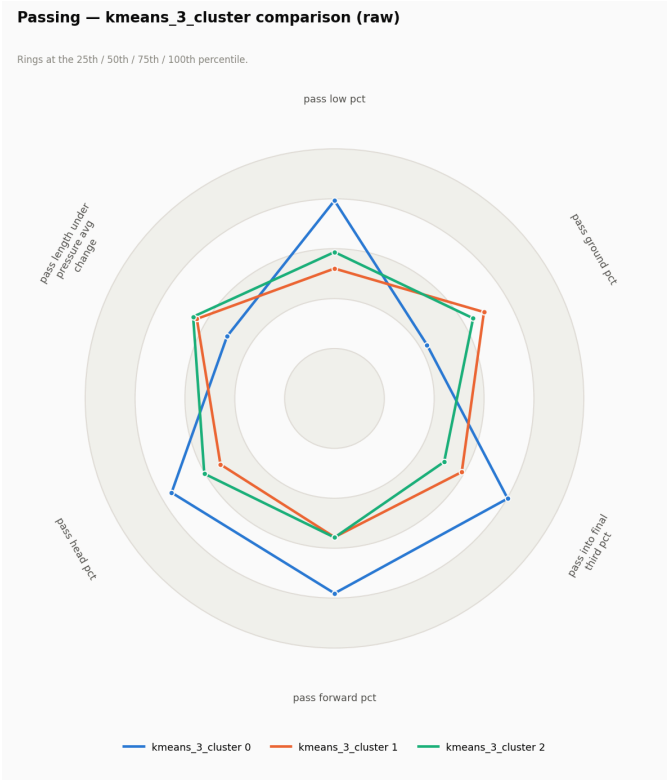


Figure 31: Center Backs Passing — Raw Statistics Comparison, 3 Clusters

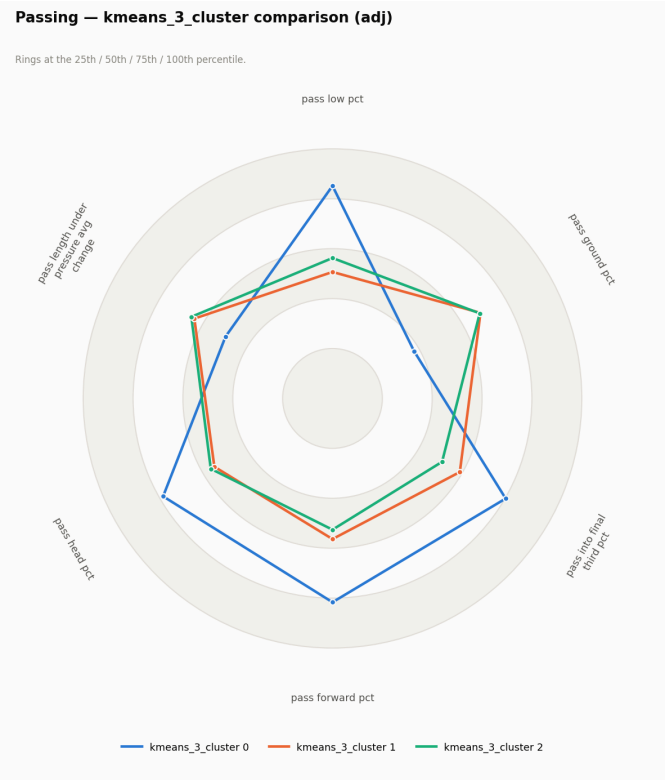


Figure 32: Center Backs Passing — Adjusted Statistics Comparison, 3 Clusters

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Carrying & Receipt

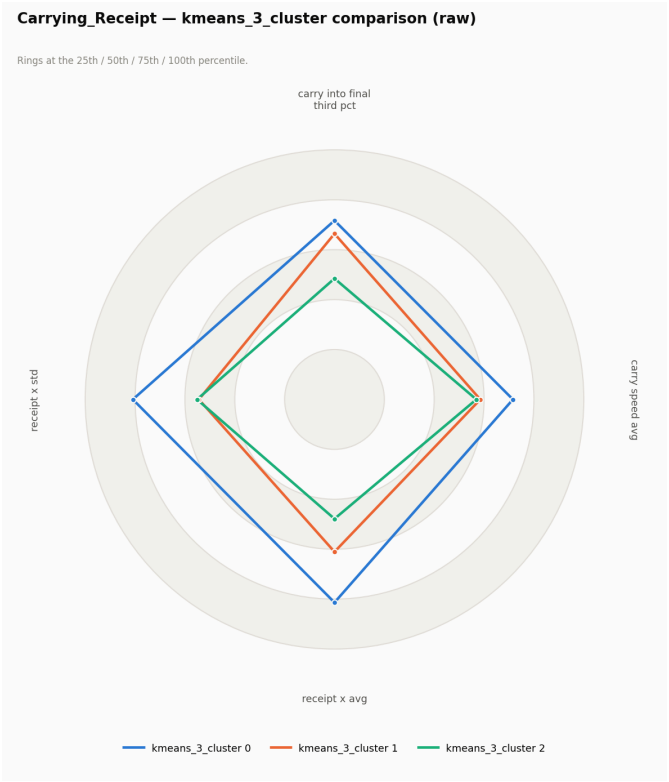


Figure 33: Center Backs Carrying Receipt — Raw Statistics Comparison, 3 Clusters

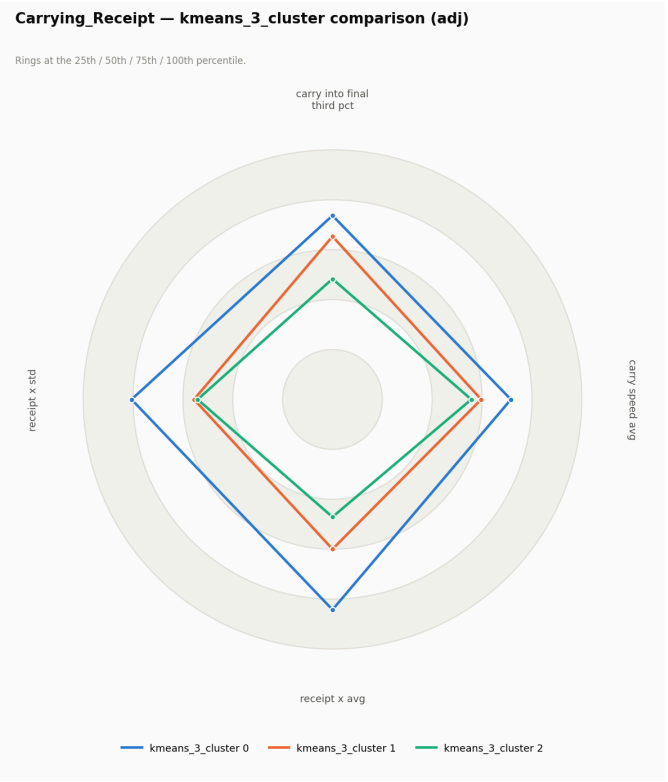


Figure 34: Center Backs Carrying Receipt — Adjusted Statistics Comparison, 3 Clusters

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Defending & Vulnerability

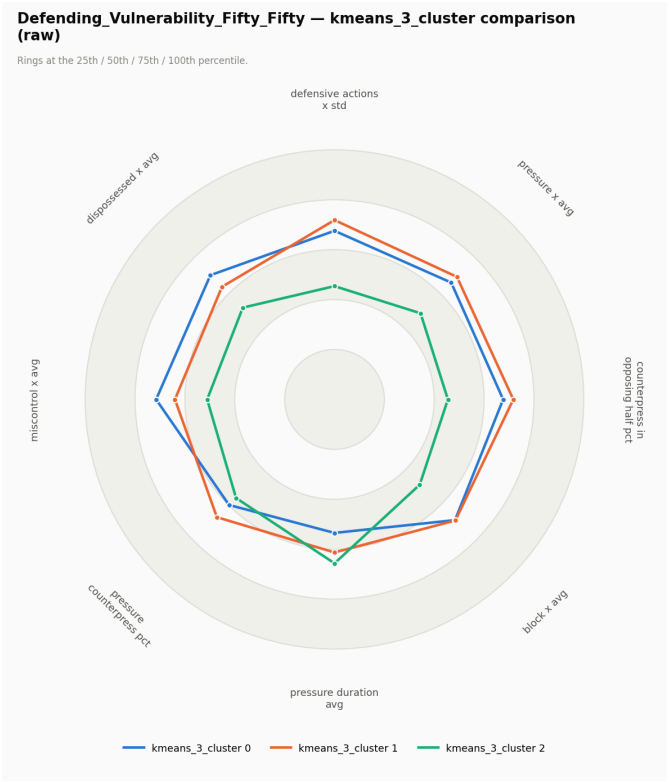


Figure 35: Center Backs Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 3 Clusters

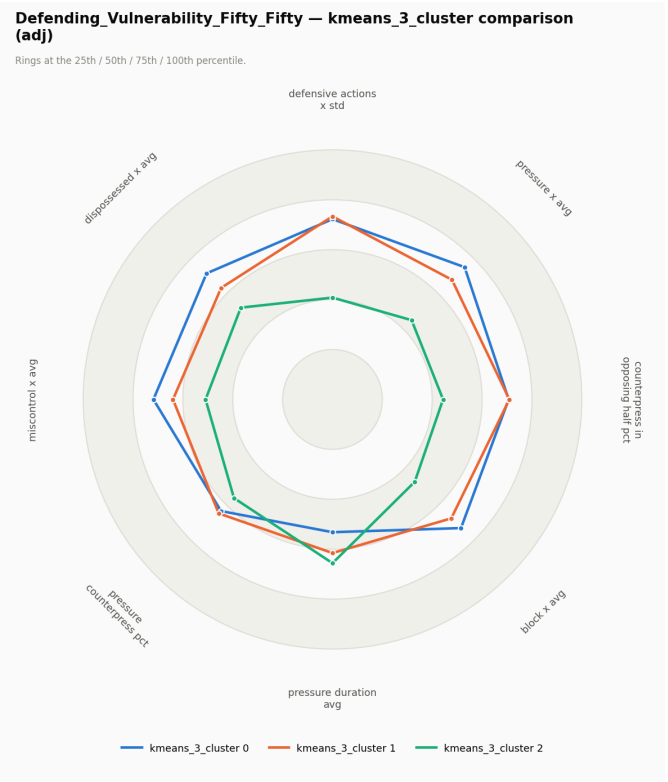


Figure 36: Center Backs Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 3 Clusters

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FA Women's Super League

Passing

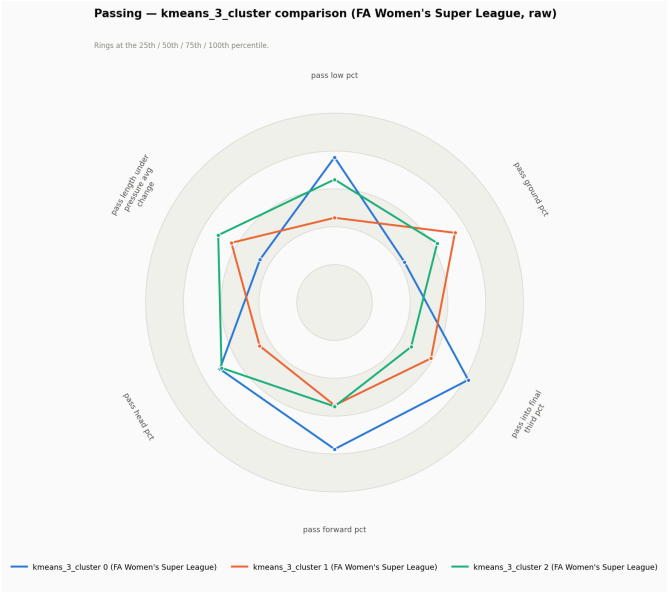


Figure 37: England Center Backs Passing — Raw Statistics Comparison, 3 Clusters

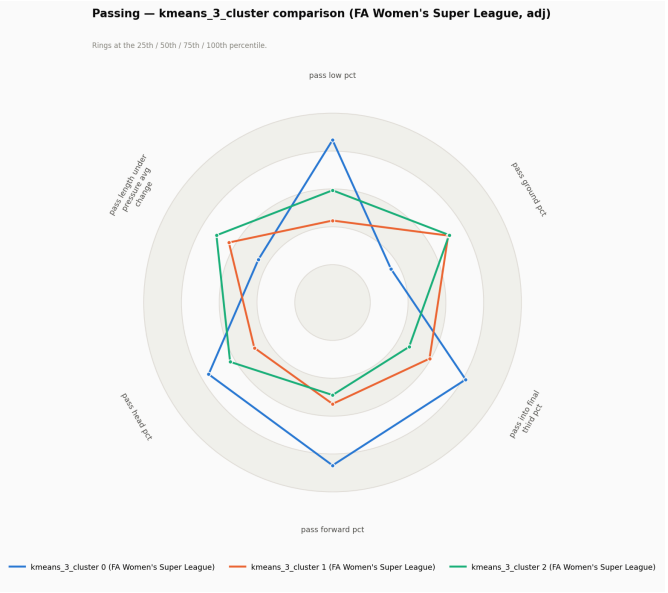


Figure 38: England Center Backs Passing — Adjusted Statistics Comparison, 3 Clusters

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Carrying & Receipt

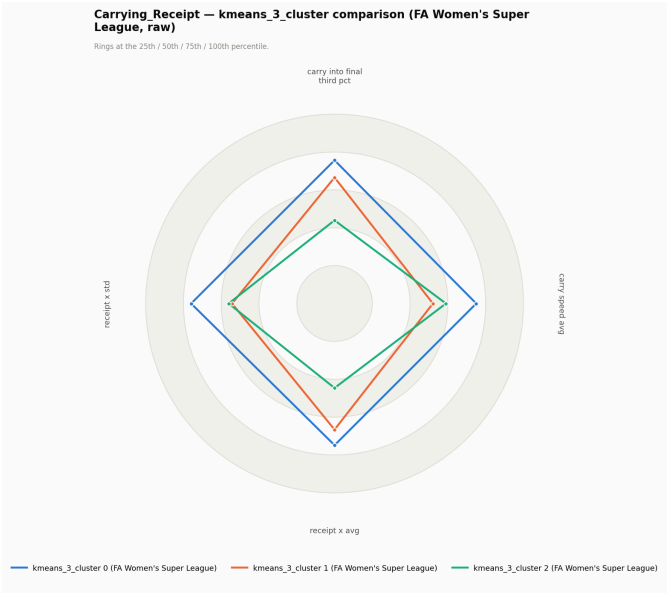


Figure 39: England Center Backs Carrying Receipt — Raw Statistics Comparison, 3 Clusters

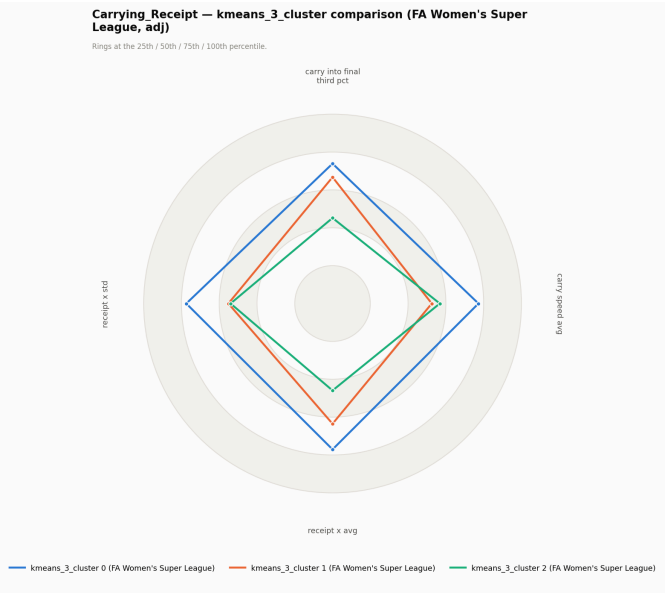


Figure 40: England Center Backs Carrying Receipt — Adjusted Statistics Comparison, 3 Clusters

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Defending & Vulnerability

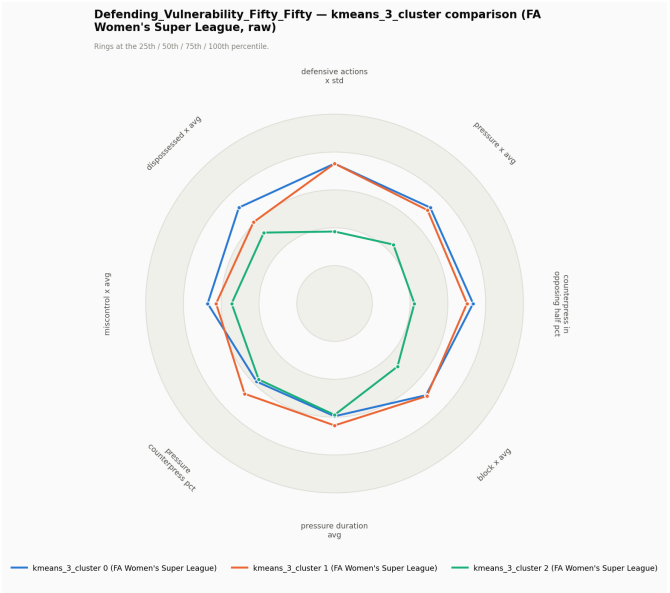


Figure 41: England Center Backs Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 3 Clusters

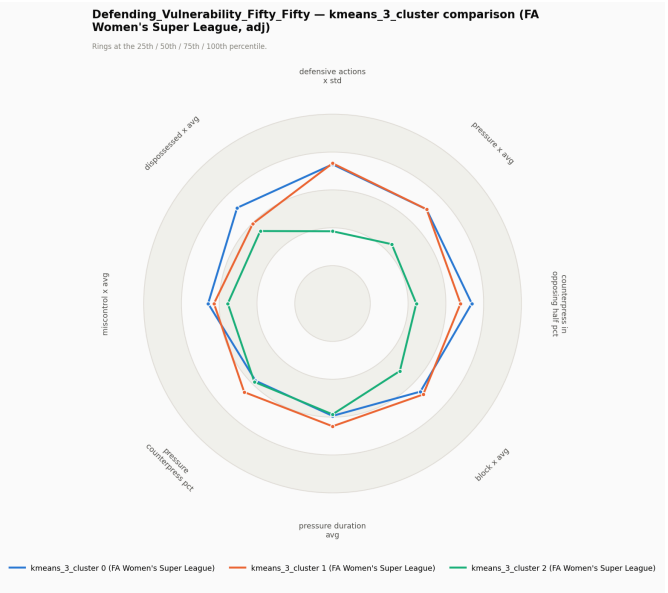


Figure 42: England Center Backs Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 3 Clusters

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Frauen Bundesliga

Passing

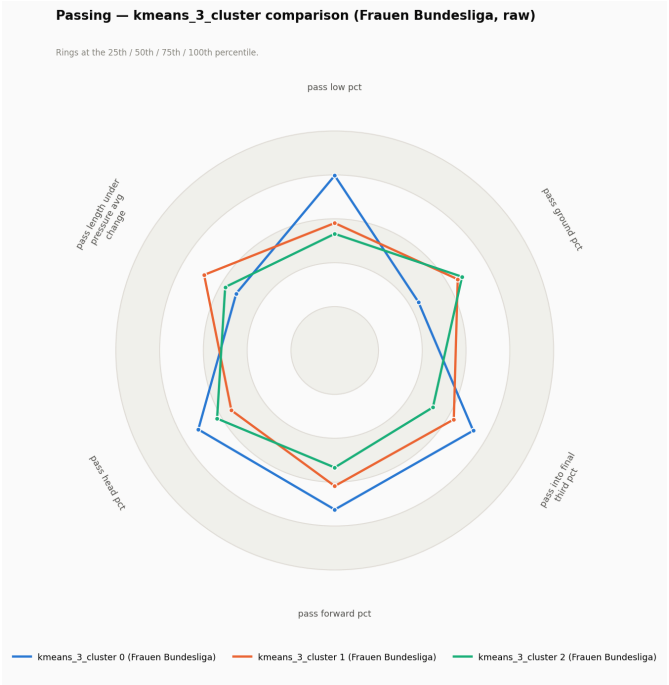


Figure 43: Germany Center Backs Passing — Raw Statistics Comparison, 3 Clusters

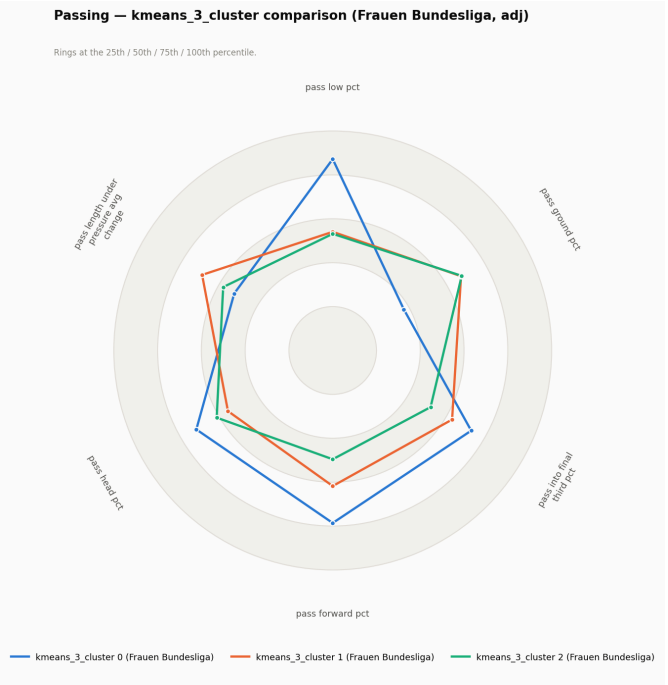


Figure 44: Germany Center Backs Passing — Adjusted Statistics Comparison, 3 Clusters

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Carrying & Receipt

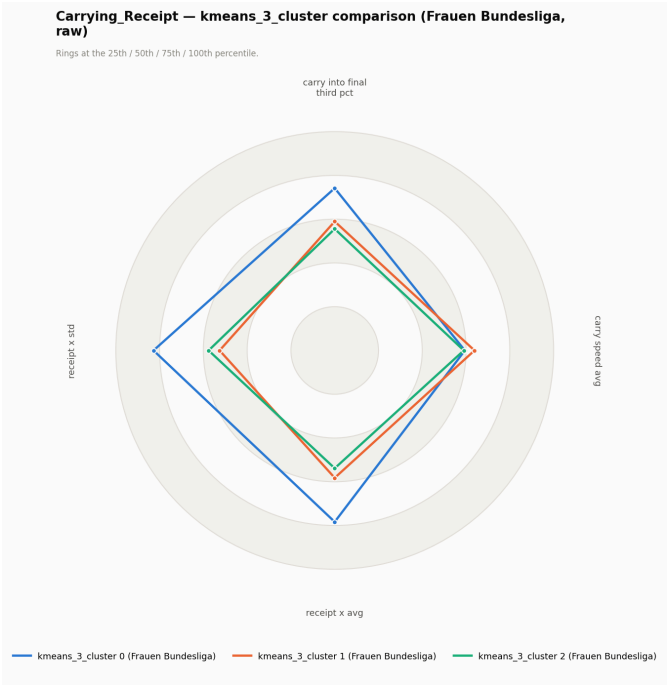


Figure 45: Germany Center Backs Carrying Receipt — Raw Statistics Comparison, 3 Clusters

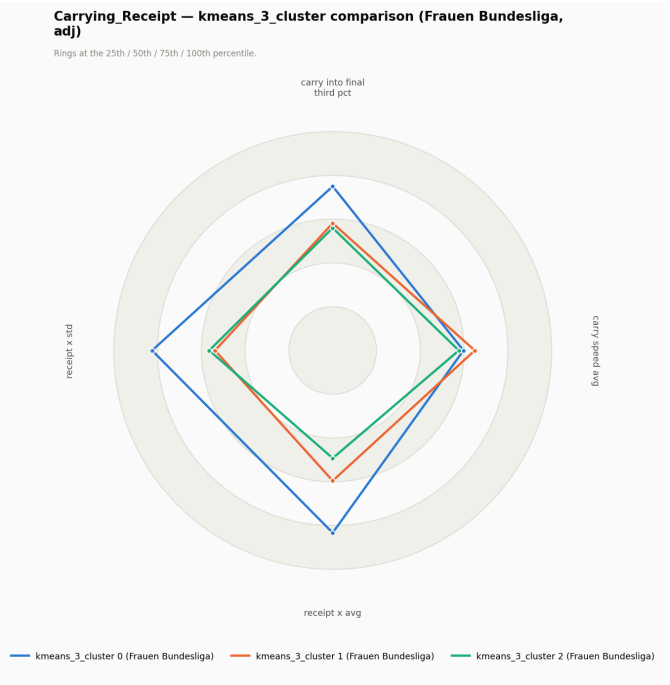


Figure 46: Germany Center Backs Carrying Receipt — Adjusted Statistics Comparison, 3 Clusters

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Defending & Vulnerability

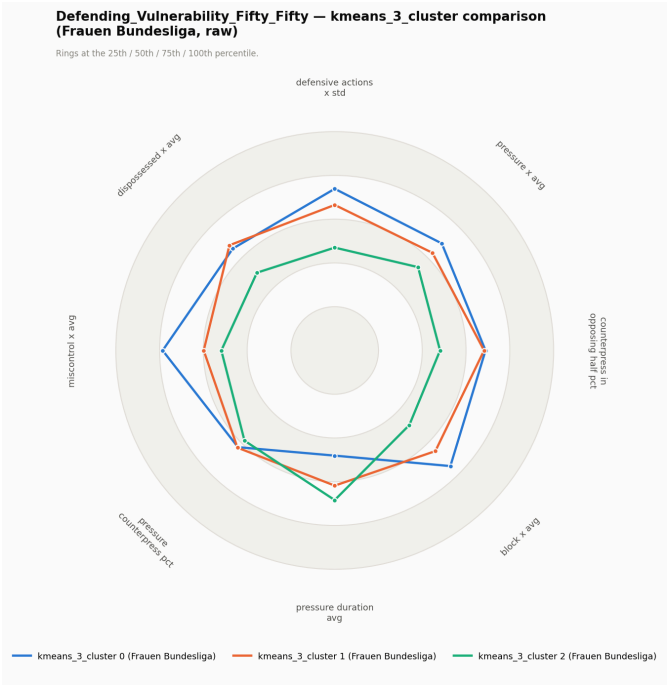


Figure 47: Germany Center Backs Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 3 Clusters

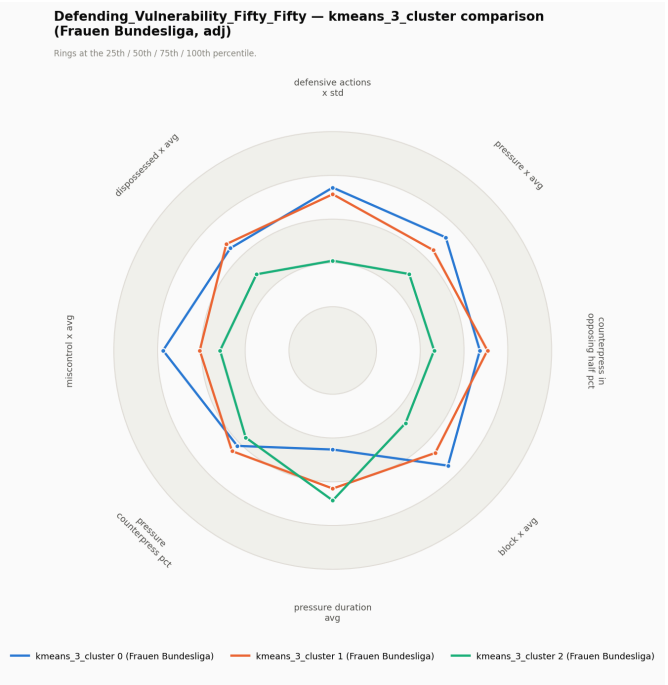


Figure 48: Germany Center Backs Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 3 Clusters

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Liga F

Passing

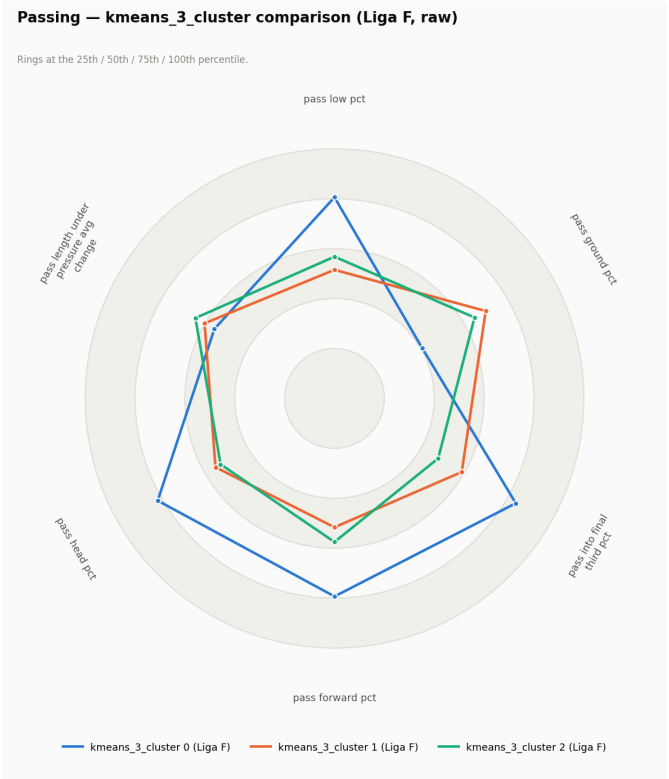


Figure 49: Spain Center Backs Passing — Raw Statistics Comparison, 3 Clusters

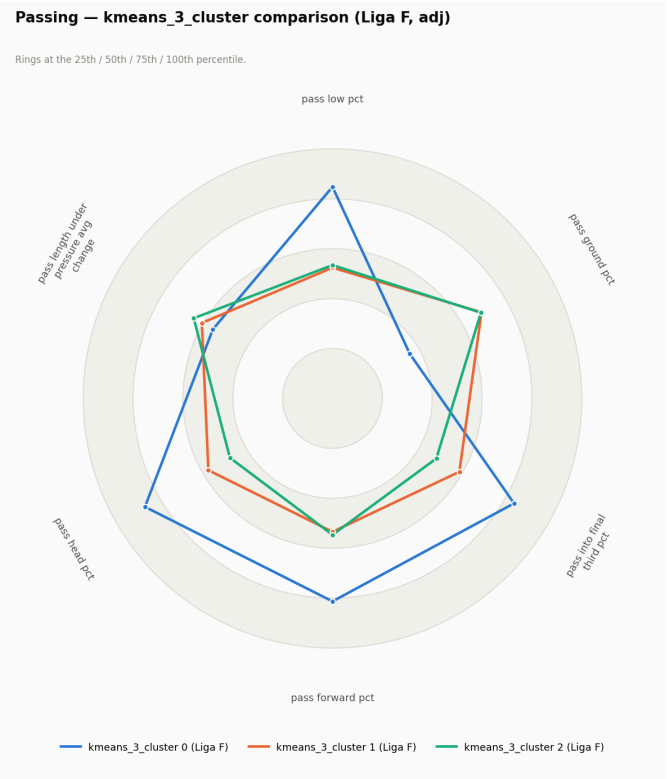


Figure 50: Spain Center Backs Passing — Adjusted Statistics Comparison, 3 Clusters

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Carrying & Receipt

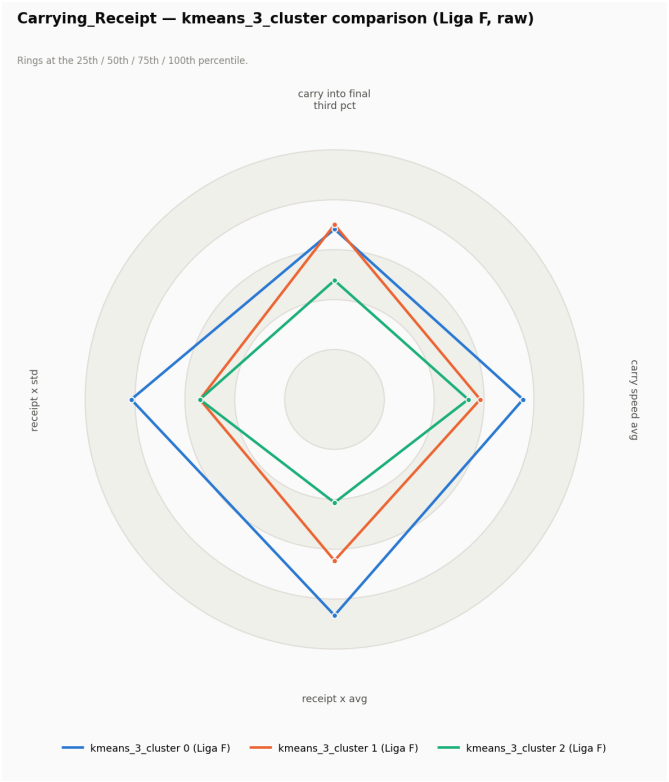


Figure 51: Spain Center Backs Carrying Receipt — Raw Statistics Comparison, 3 Clusters

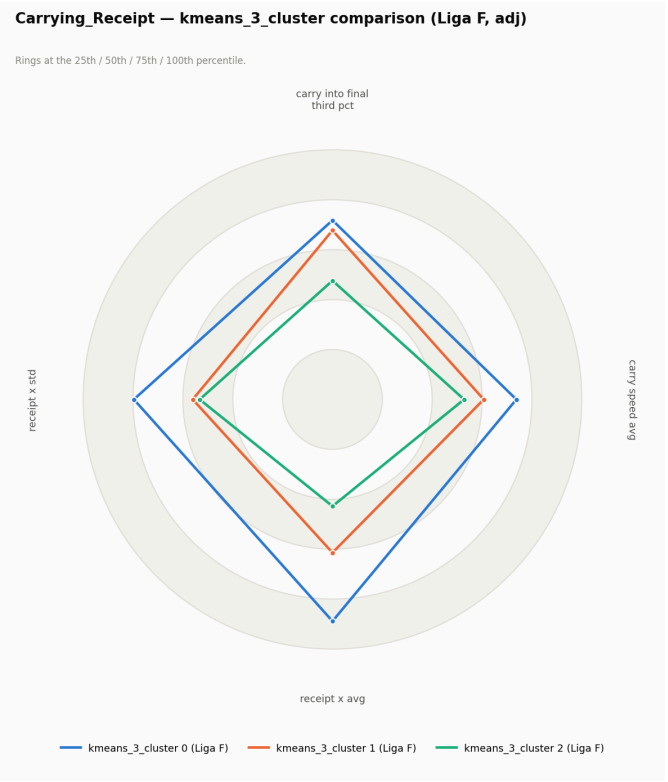


Figure 52: Spain Center Backs Carrying Receipt — Adjusted Statistics Comparison, 3 Clusters

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Defending & Vulnerability

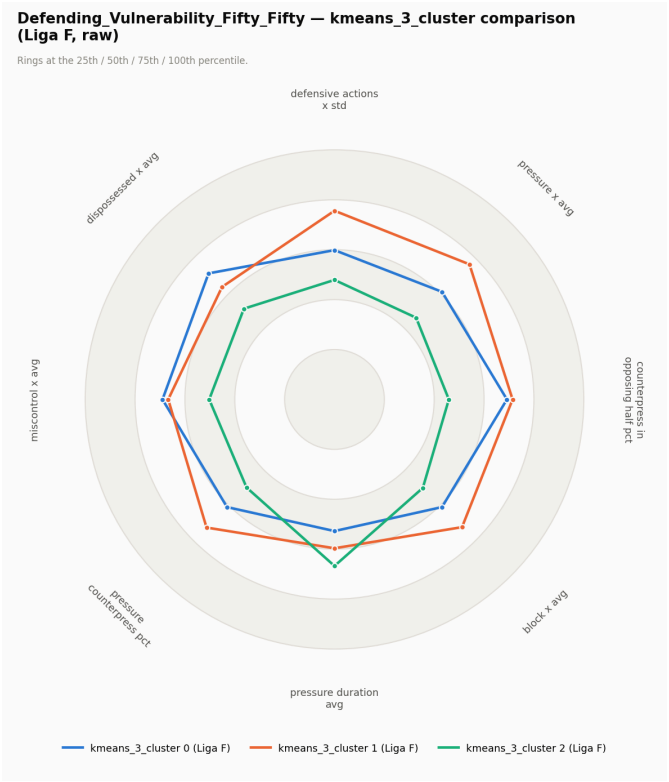


Figure 53: Spain Center Backs Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 3 Clusters

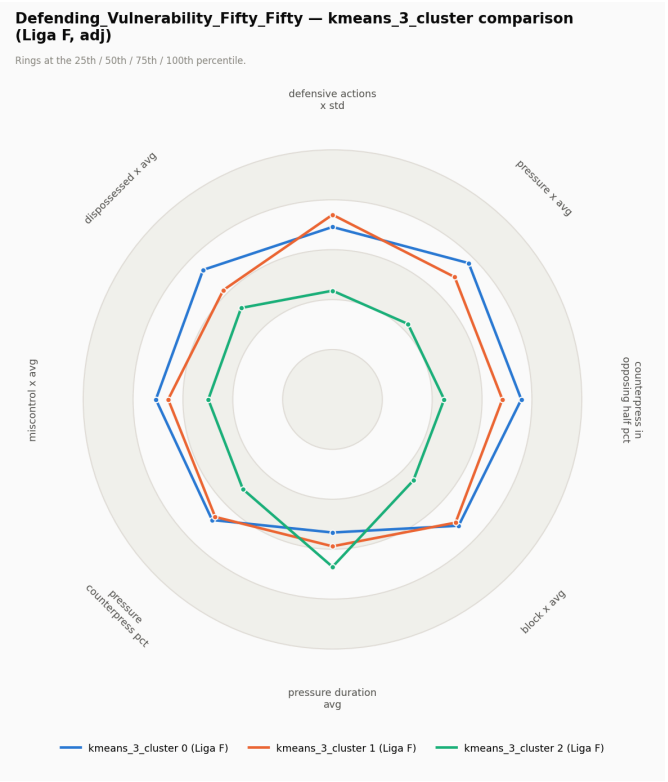


Figure 54: Spain Center Backs Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 3 Clusters

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Serie A Women

Passing

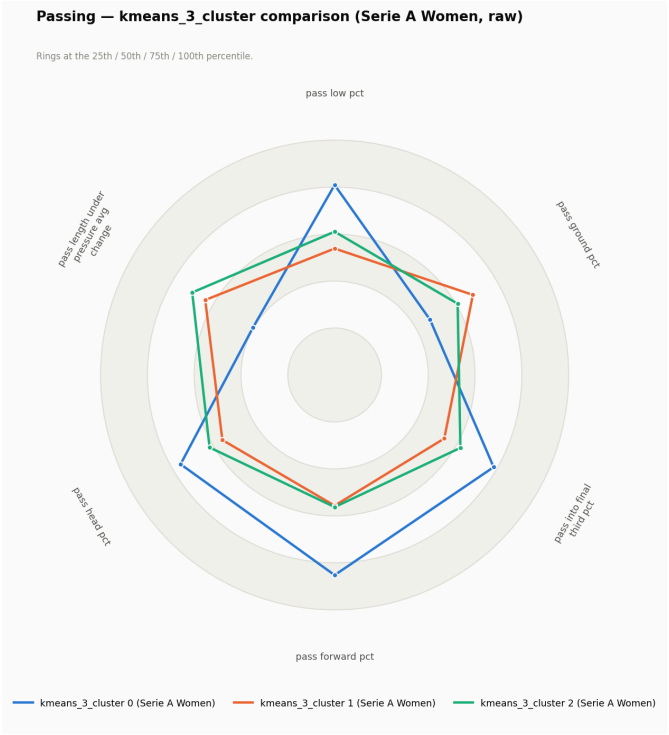


Figure 55: Italy Center Backs Passing — Raw Statistics Comparison, 3 Clusters

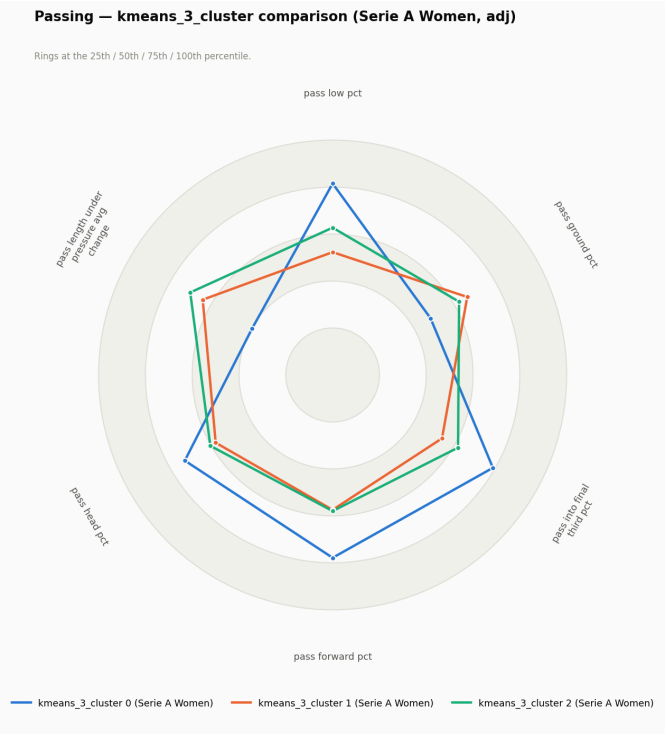


Figure 56: Italy Center Backs Passing — Adjusted Statistics Comparison, 3 Clusters

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Carrying & Receipt

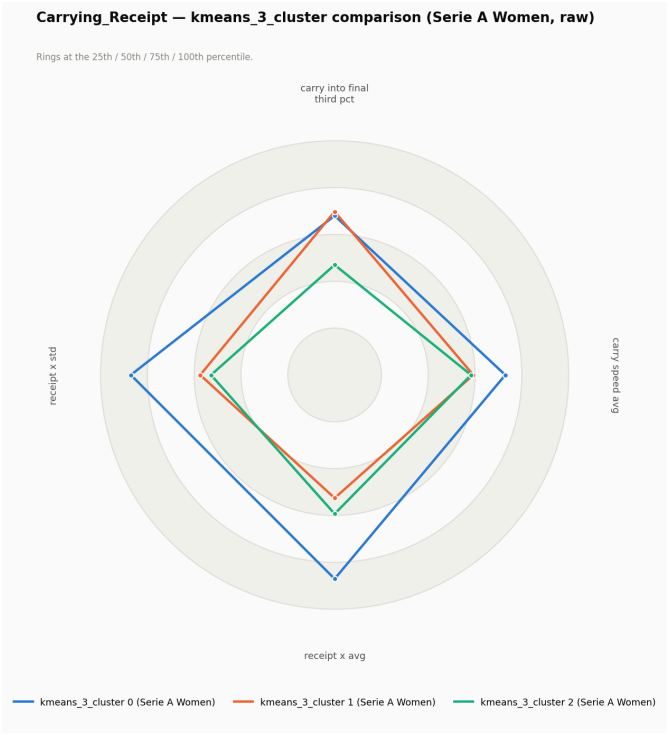


Figure 57: Italy Center Backs Carrying Receipt — Raw Statistics Comparison, 3 Clusters

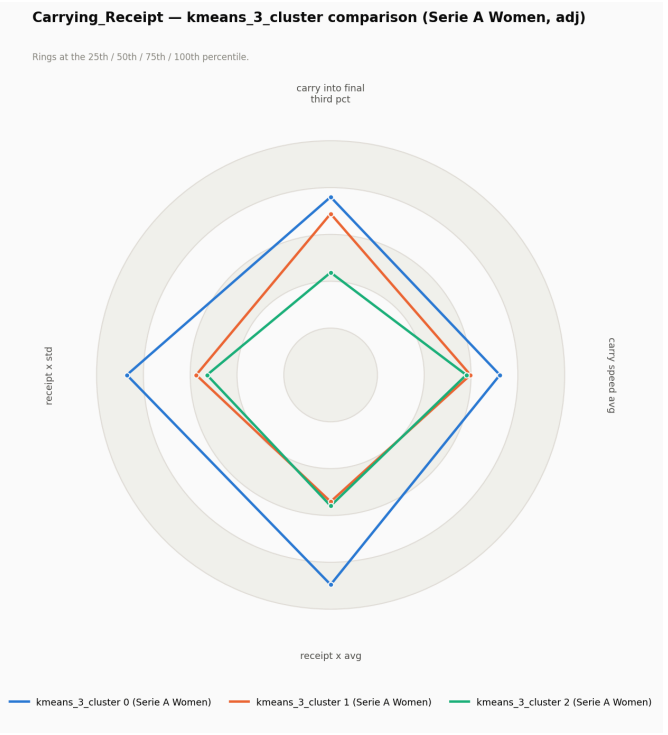


Figure 58: Italy Center Backs Carrying Receipt — Adjusted Statistics Comparison, 3 Clusters

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Defending & Vulnerability

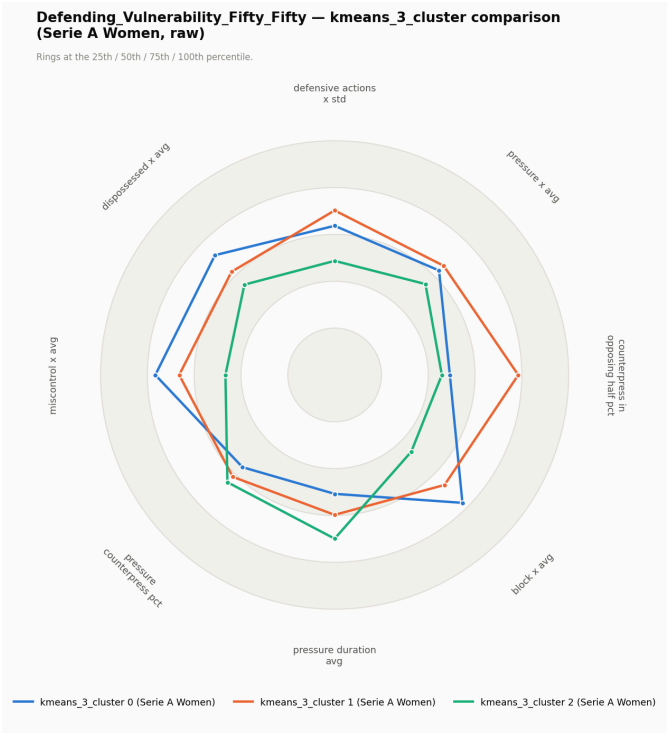


Figure 59: Italy Center Backs Defending Vulnerability Fifty Fifty — Raw Statistics Comparison, 3 Clusters

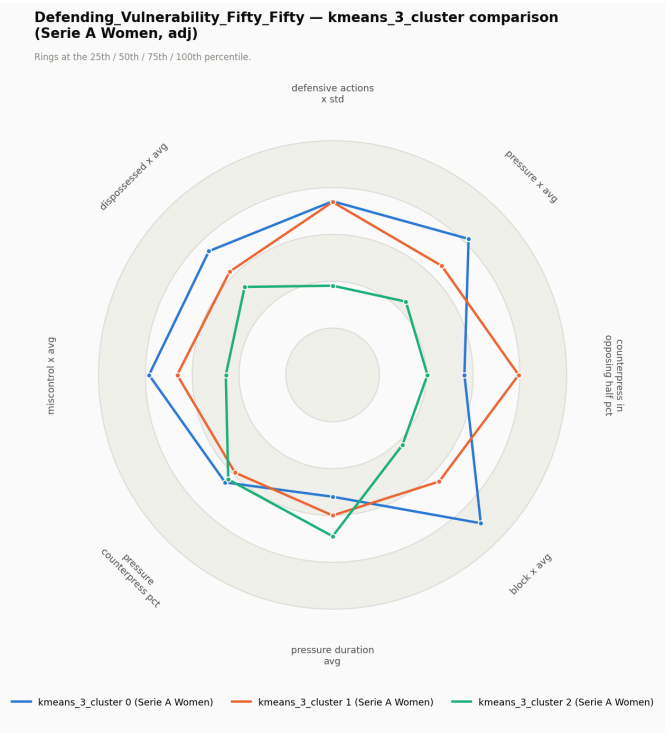


Figure 60: Italy Center Backs Defending Vulnerability Fifty Fifty — Adjusted Statistics Comparison, 3 Clusters